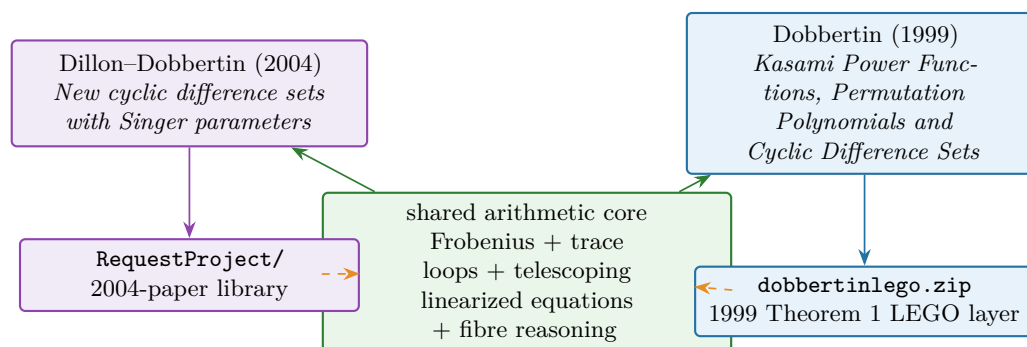


The Dobbertin LEGO Atlas

Building blocks shared by two papers and their Lean formalizations



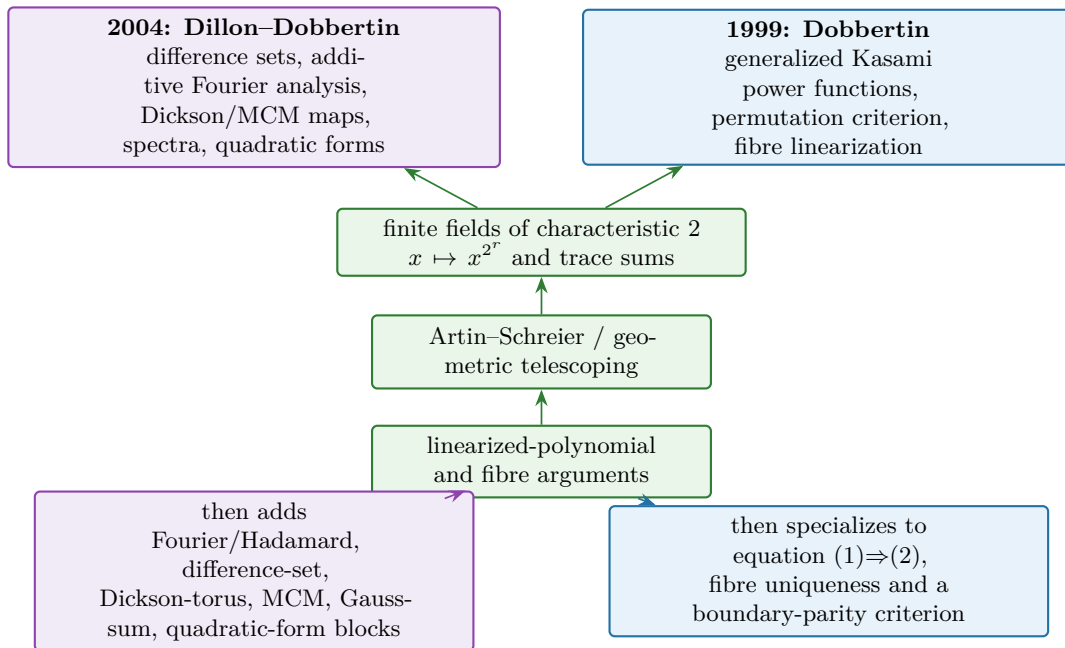
Short answer. The projects share a real and reusable lower layer, but they are *not* the same construction. The ZIP is centered on Theorem 1 of a 1999 Dobbertin paper; the uploaded paper and the `RequestProject` library follow the much broader 2004 Dillon–Dobbertin paper. Frobenius, trace/partial-trace loops, telescoping, linearized polynomials, and fibre uniqueness can be built from essentially the same bricks. Hadamard analysis, Singer difference sets, Dickson tori, MCM–Dickson identities, characteristic-two quadratic forms, cubic characters/Gauss sums, and design inequivalence require further brick families already represented in `RequestProject/Foundations`.

Colour code: ■ shared ■ adaptable/partial ■ 2004-specific families ■ 1999-LEGO presentation.

Page references below use the printed journal pagination; the parenthesized number is the page in the supplied 48-page PDF.

1. First separate the two source papers

Two related mathematical stories, one common arithmetic engine



What the ZIP actually contains

The archive’s top-level documentation explicitly names the 1999 paper and describes the path

Frobenius → Loop → Assembly → FiberReduction → FiberUniqueness → Theorem 1.

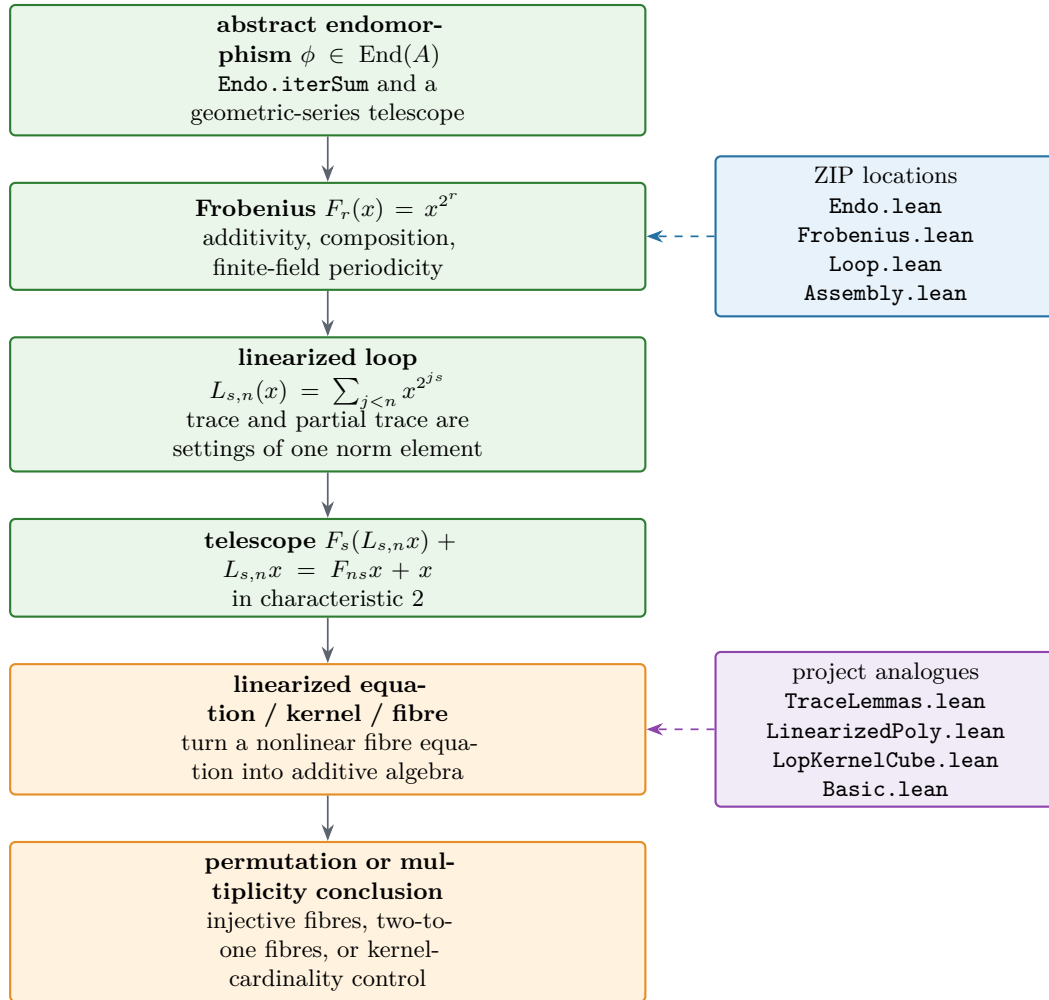
Its self-contained generic core is concentrated in `Endo.lean`, `Frobenius.lean`, `Loop.lean`, and `Assembly.lean`; it also supplies algebraic, categorical, object-first, and Mac Lane presentations of that core.

Packaging caveat. The archive is not, by itself, the complete 1999 formalization: `FiberReduction.lean`, `FiberUniqueness.lean` and `Theorem1.lean` import `Dobbertin.Theorem1.*` modules that are not present in the ZIP. Thus the generic LEGO core is reusable, while the end-to-end facade expects another library to be available.

What the current project contains

`RequestProject/` is a modular transcription and growing proof library for the 2004 paper. Its paper-facing files mirror Sections 1–8 and Appendix A; its `Foundations/` directory contains the bottom-up reusable infrastructure. Some declarations remain explicit `sorry` boxes, so the atlas distinguishes *available bricks and interfaces* from a fully completed proof.

2. The common LEGO spine

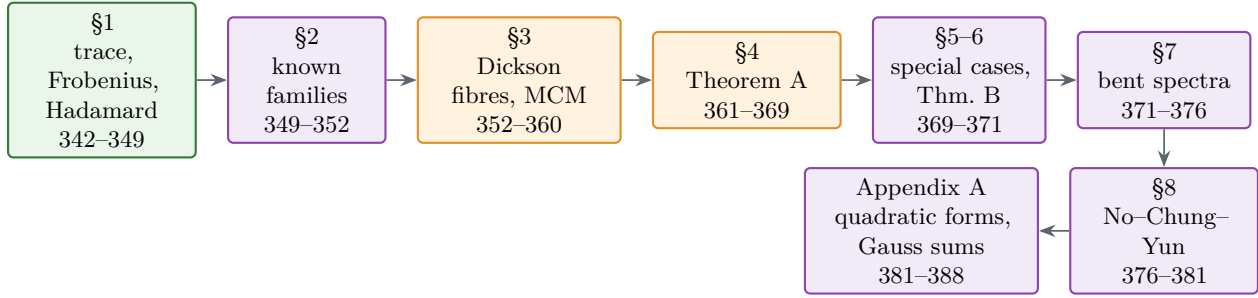


Brick	dobbertinlego.zip	2004 formalization / mathematical use
Endomorphism norm	iterSum, iterSum_telescope, normElt	General pattern behind trace loops; reusable in LinearizedPoly and trace identities.
Frobenius	frob, frobEndo, periodicity	Field powers and trace invariance throughout; explicit support in TraceLemmas.
Trace loop	loop, trace, partialTrace, numeratorSum	Absolute/relative traces in Sections 1, 4, 7, 8 and Appendix A.
Linearization	equation (1) $\Rightarrow$ affine Frobenius equation (2)	MCM telescoping, $L_{k,a}$ kernels, quadratic-form radicals.
Fibres	uniqueness and boundary collisions	multiplicity functions, 2-to-1 maps, Dickson fibres, kernel-size arguments.

The correct reuse claim is structural, not literal. The same abstract telescope can power both developments. The concrete equations, target maps, and final theorems differ, so an adapter layer is needed rather than a wholesale import-and-finish operation.

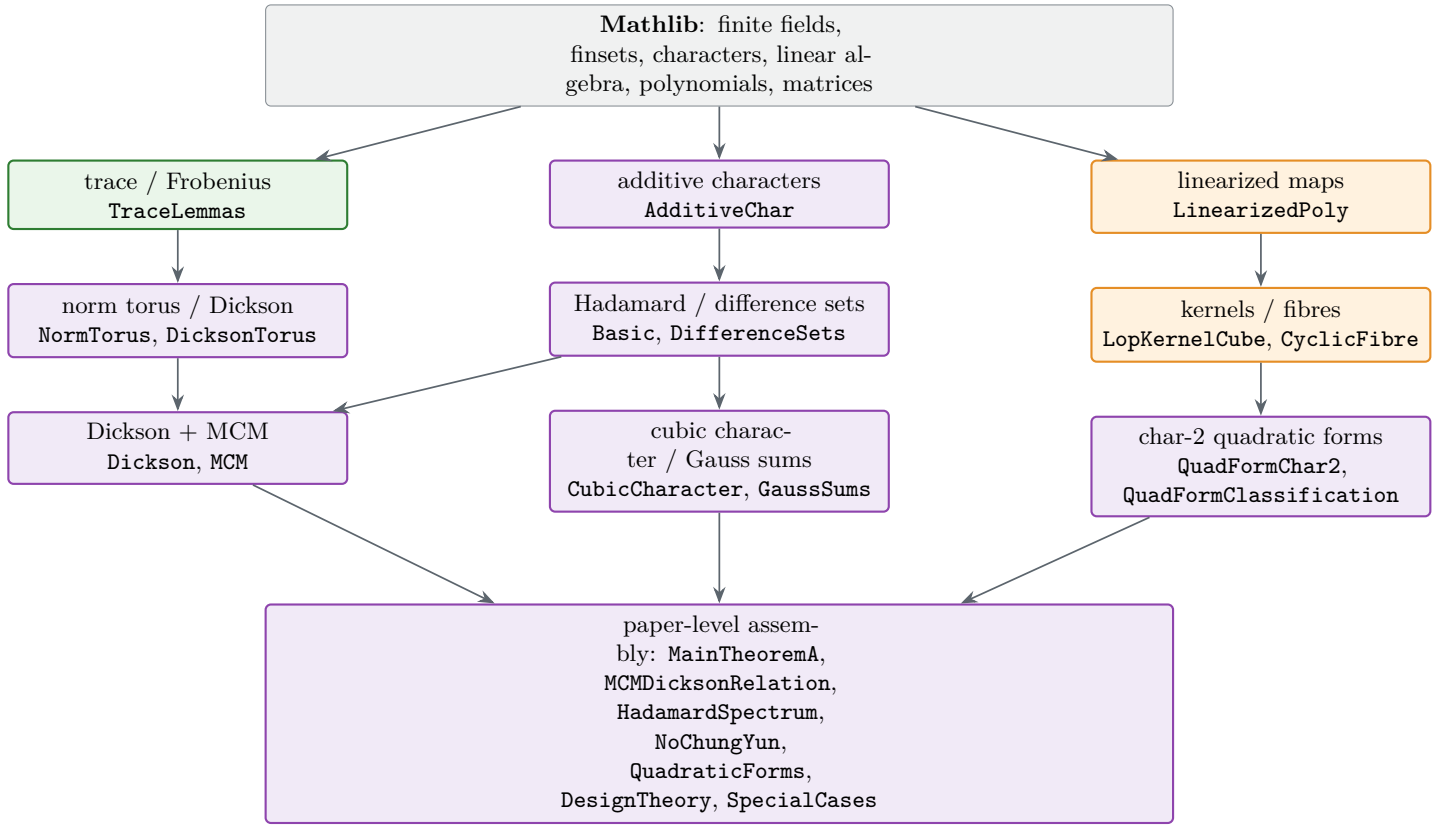
3. Where the blocks occur in the 2004 paper

Paper route: printed pages 342–389 (PDF pages 1–48)



Paper place	PDF page(s)	Mathematical blocks visible there	Main Lean destination
§1, pp. 342–349	1–8	$L = \text{GF}(2^m)$, trace character, normalized Hadamard transform, convolution, Parseval, Singer difference-set criteria	Basic, DifferenceSets, AdditiveChar
§3, pp. 352–360	11–19	Dickson polynomial identity and permutation criterion (Thm. 3); fibre sizes (Prop. 5); multiplicity (Prop. 6, Thm. 7); MCM maps (Thms. 8–10)	Dickson, MCM, DicksonTorus, CyclicFibre, LinearizedPoly
§4, pp. 361–369	20–28	Theorem A; Lemmas A1/A2; trace substitutions and Fourier/Hadamard equivalence	MainTheoremA, GaussSumTheory, QuadFormChar2
§5–6, pp. 369–371	28–30	special cases; MCM–Dickson transform dictionary (Theorem B)	SpecialCases, MCMDicksonRelation
§7, pp. 371–376	30–35	bent functions and three-valued spectra (Thm. 11, Cor. 12)	HadamardSpectrum, QuadFormChar2
§8, pp. 376–381	35–40	No–Chung–Yun difference set (Thms. 13, 15 and corollaries)	NoChungYun, DesignTheory
App. A, pp. 381–388	40–47	rank/radical, canonical forms, Fourier spectrum of quadratic forms, trace forms, cubic Gauss sum	QuadraticForms, QuadFormClassification, GaussSums, CubicCharacter

4. The 2004 formalization as a LEGO city



Reading the colours. The green and orange columns are the best candidates for direct conceptual reuse from the ZIP. Purple nodes are not dispensable decoration: they encode the extra mathematics by which the 2004 paper turns finite-field maps into cyclic difference sets, Fourier spectra, and inequivalence results.

Family	Input supplied by the common spine	New connector required in the 2004 library
Trace/Frobenius	additivity, iteration, periodicity, norm-element telescope	identify Mathlib field trace and the project's <code>Tr</code> ; relative-trace towers
Linearized maps	loops and affine Frobenius equations	kernel dimensions, fibre cardinalities, $L_{k,a}$ and <code>Lop</code> identities
Dickson fibres	generic finite-map fibre reasoning	split/norm-one tori of orders $q - 1, q + 1$ and $z + z^{-1}$ folding
MCM maps	partial-trace telescopes	exact $f_k, Q_{k,k'}, R_{k,k'}$ formulas and transform identity
Quadratic forms	Frobenius and trace algebra	polar forms, radicals, rank, Arf/canonical classification, spectrum laws
Fourier/difference sets	essentially none beyond the additive character's trace	character orthogonality, Hadamard involution/convolution, difference-set equivalences

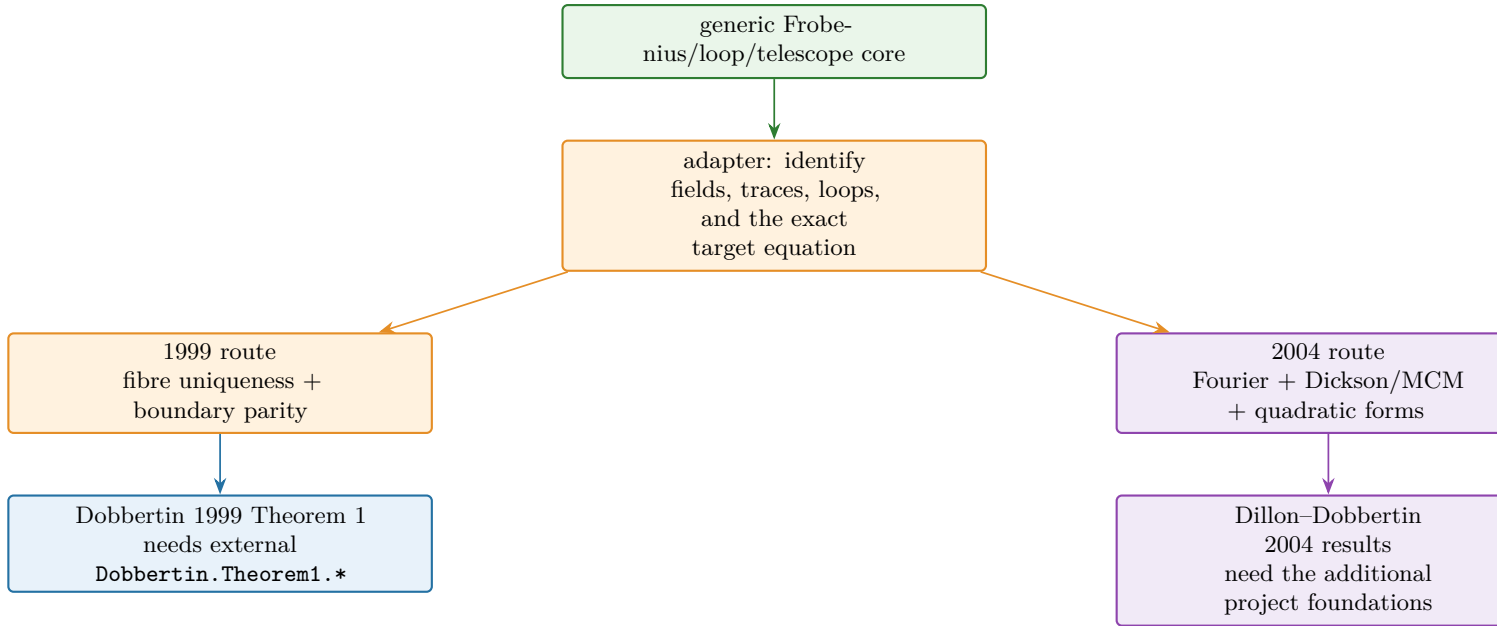
5. Cross-paper compatibility matrix

Brick family	Reuse verdict	ZIP source	2004 paper / project destination
Abstract geometric sum in $\text{End}(A)$	common	<code>Endo.iterSum</code> , <code>iterSum_telescope</code> ; categorical variants in <code>Categorical</code> , <code>CategoryTheory</code> , <code>MacLaneLadder</code>	Underlying pattern for trace and linearized sums; closest destinations: <code>Foundations/TraceLemmas</code> , <code>LinearizedPoly</code>
Frobenius x^{2^r}	common	<code>Frobenius.frob</code> , <code>frobEndo</code> ; generalized in <code>GenBase.baseFrob</code>	Used throughout 2004; explicit in <code>TraceLemmas</code> , <code>LinearizedPoly</code> , <code>QuadFormChar2</code>
Trace and partial trace	common	<code>Loop.loop</code> ; wrappers in <code>Assembly.trace</code> , <code>partialTrace</code>	§1 and later trace substitutions; <code>Basic.Tr</code> , <code>TraceLemmas</code> , <code>Constructions</code>
Artin–Schreier telescope	common	<code>loop_telescope</code> , <code>partialTrace_telescope</code>	MCM telescope and kernels; <code>LinearizedPoly.MCM_telescope</code> , trace/radical computations
Affine linearized equation	partial	<code>Assembly.linearized</code> ; equation $(1) \Rightarrow (2)$	Same technique, but different equations: MCM maps, $L_{k,a}$, and trace quadratic forms
Fibre uniqueness/cardinality	partial	<code>FiberReduction</code> , <code>FiberUniqueness</code> (requires external 1999 modules)	Prop. 5, Thms. 7/10; <code>CyclicFibre</code> , <code>Basic.mult</code> , <code>Dickson</code>
Permutation by boundary parity	1999-specific	<code>Theorem1</code> facade	Not the organizing theorem of the 2004 paper
Hadamard/Fourier analysis	additional	absent	§1, §4, §6–8; <code>AdditiveChar</code> , <code>Basic</code> , <code>MCMDicksonRelation</code>
Singer difference sets/designs	additional	only in the 1999 paper’s broader title/context, not a ZIP brick family	Definition 1, Prop. 2, Theorem A, Thm. 13; <code>DifferenceSets</code> , <code>DesignTheory</code>
Dickson split/norm tori	additional	absent	§3 Thm. 3/Prop. 5; <code>NormTorus</code> , <code>DicksonTorus</code> , <code>Dickson</code>
MCM inverse/transform theory	partial	trace-loop technology only	§3 Thms. 8–10 and §6 Thm. B; MCM, <code>MCMDicksonRelation</code>
Quadratic-form classification	additional	absent	Appendix A and §7–8; <code>QuadFormChar2</code> , <code>QuadFormClassification</code> , <code>QuadraticForms</code>
Cubic characters/Gauss sums	additional	absent	Appendix A and §4/7/8; <code>CubicCharacter</code> , <code>GaussSums</code> , <code>GaussSumTheory</code>
Inequivalence / binary rank	additional	absent	Theorem A inequivalence; <code>DesignTheory</code>

Largest genuine intersection. Both developments repeatedly exploit the fact that in characteristic two, Frobenius is additive and a sum of its iterates telescopes. This turns nonlinear-looking power equations into additive or linearized equations whose kernels and fibres can be counted. That is the common LEGO grammar.

6. A precise buildability picture

What can be assembled from the ZIP bricks alone?



Four answers to “can be built from the same blocks?”

1. **The generic trace/telescope layer: yes.** The ZIP deliberately abstracts this layer farther than the paper-facing project, including commutative-algebra and categorical versions.
2. **The 2004 linearized-polynomial subarguments: often, after adapters.** Definitions and exponent conventions must be matched, and the target kernel/fibre theorem still has to be proved.
3. **The complete 2004 paper: no, not from those blocks alone.** Its Fourier, difference-set, Dickson-torus, quadratic-form, Gauss-sum, and design-theory layers are independent additions.
4. **The complete ZIP facade: not from the archive alone.** Its last three modules refer to unbundled `Dobbertin.Theorem1.*` dependencies. This does not invalidate the generic core, but it prevents treating the ZIP as a standalone complete proof package.

Recommended integration boundary

Do not merge the two theorem namespaces wholesale. Extract or import only a small neutral layer:

`iterSum` `Frob` `loop` `telescope` `finite-order/fixed-point lemmas`.

Then prove project-local bridge lemmas to `RequestProject.Tr`, partial traces, `MCM_telescope`, and the relevant linearized maps. This preserves discoverability and avoids forcing the 2004 library through the ZIP’s 1999-specific equations or categorical facades.

7. File-finder: where to look first

Question	First ZIP file(s)	First project / paper location
Where is the abstract telescope?	DobbertinLego/Endo.lean	Foundations/LinearizedPoly.lean; conceptual uses in 2004 §3–4
Where is Frobenius made additive?	Frobenius.lean; general base in GenBase.lean	Foundations/TraceLemmas.lean; used from 2004 §1 onward
Where are trace sums one loop?	Loop.lean, Assembly.lean	Basic.lean, TraceLemmas.lean; 2004 p. 344 (PDF 3)
Where is equation linearization?	Assembly.lean, AlgebraLayer.lean	LinearizedPoly.lean, LopKernelCube.lean; 2004 §3 and Appendix A
Where are finite fibres counted?	FiberReduction.lean, FiberUniqueness.lean	Basic.mult, CyclicFibre.lean, Dickson.lean; Prop. 5 pp. 353–354 (PDF 12–13)
Where is additive Fourier theory?	not in the ZIP	AdditiveChar.lean, Basic.lean; 2004 pp. 344–345 (PDF 3–4)
Where are difference sets?	no dedicated ZIP module	DifferenceSets.lean; Definition 1/Prop. 2 pp. 345–347 (PDF 4–6)
Where is Dickson theory?	not in the ZIP	NormTorus.lean, DicksonTorus.lean, Dickson.lean; 2004 §3
Where is MCM theory?	only generic trace tools	MCM.lean, MCMDicksonRelation.lean; Thms. 8–10 pp. 359–360 (PDF 18–19), Thm. B p. 370 (PDF 29)
Where are quadratic forms?	not in the ZIP	QuadFormChar2.lean, QuadFormClassification.lean, QuadraticForms.lean; Appendix A pp. 381–388 (PDF 40–47)
Where are cubic Gauss sums?	not in the ZIP	CubicCharacter.lean, GaussSums.lean, GaussSumTheory.lean; Prop. A.4 pp. 386–387 (PDF 45–46)
Where is final assembly?	Theorem1.lean (1999, with external imports)	MainTheoremA.lean, HadamardSpectrum.lean, NoChungYun.lean (2004)



Bottom line: one reusable arithmetic chassis, two different mathematical vehicles.